

Mohamed Sameh Salah Eldine

Data Scientist | Machine Learning Engineer | Backend Developer

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Professional Summary

Data Scientist and Backend Developer with hands-on experience building machine learning systems, AI-powered applications, OCR platforms, analytics solutions, and scalable backend architectures. Skilled in Python, SQL, PyTorch, FastAPI, Node.js, React, and modern data engineering workflows. Experienced in designing end-to-end solutions from data collection and model training to deployment, APIs, dashboards, and production-ready software systems.

Education

Faculty of Computers and Data Science, Alexandria University Sep 2022 – Jun 2026
Bachelor in Computing and Data Science GPA: **3.50 / 4.00**
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Mining, Database Systems, Cloud Computing, Data Structures, Algorithms, Operating Systems, Computer Networks, Web Development.

Technical Skills

Programming Languages: Python, SQL, TypeScript, JavaScript, C++, Java
Machine Learning & AI: PyTorch, Scikit-learn, Deep Learning, Transformers, LSTM, Computer Vision, OCR, NLP, Feature Engineering, Hyperparameter Optimization
Data Science: Pandas, NumPy, Statistical Analysis, Predictive Modeling, Data Cleaning, Data Visualization
Data Engineering: ETL Pipelines, PostgreSQL, MySQL, SQL Server, Database Design, Data Processing
Backend Development: FastAPI, Node.js, Express.js, REST APIs, JWT Authentication, OAuth, Microservices Architecture
Frontend & Visualization: React, Streamlit, Power BI, Plotly, Matplotlib, Seaborn
Cloud & DevOps: Docker, GitHub Actions, CI/CD, Redis, BullMQ, Git, Linux

Experience

IoT Field Training Jul 2024 – Sep 2024

- Designed and implemented a Smart Garage System using ESP32 and Arduino.
- Developed a Flutter mobile application for monitoring and activity tracking.
- Integrated MQTT and Firebase for cloud communication and real-time synchronization.

Data Science Training – MAIM Digital Solutions Sep 2025

- Completed practical training covering machine learning workflows, data analytics, and real-world project implementation.

Projects

AI Candlestick Trader PyTorch, Transformers, LSTM, Optuna

- Developed a deep-learning trading system for EGX and Tadawul stock markets.
- Engineered 22 technical indicators and 18 candlestick pattern detectors.
- Built Transformer and Attention-based Bi-LSTM models with ensemble learning.
- Implemented Optuna hyperparameter optimization and walk-forward backtesting.
- Developed an interactive Streamlit dashboard with Plotly visualizations.

Hasibha – Personal Finance & Productivity Backend System Node.js, Express.js, MongoDB
Graduation Project | Faculty of Computers and Data Science, Alexandria University | Role: Backend Developer
(Currently under active development – pending Google Play Store release)

- Designed and developed the complete backend architecture for *Hasibha*, an Arabic-first personal finance and productivity platform targeting Egyptian users, using Node.js, Express.js, and MongoDB.
- Built a secure, scalable, and modular backend consisting of two independent layers: **auth-backend** (authentication service) and **home-backend** (core business logic), improving maintainability and enabling future scaling.
- Implemented a robust JWT authentication system with email verification, refresh token rotation, rate limiting, and comprehensive email logging, ensuring high security standards for handling sensitive financial data.
- Developed multiple core backend modules: Transactions, Saving Goals with progress tracking, Multi-Account Management, Document Storage, Dashboard Analytics, and Profile & Settings.
- Enforced strict data ownership and privacy by implementing user-scoped queries (`req.user.id`) across all modules to prevent unauthorized data access.
- Utilized MongoDB Aggregation Pipeline to build an efficient financial statistics and reporting engine.
- Applied industry best practices: MVC pattern, middleware architecture, input validation, centralized error handling,

and security hardening (bcrypt, Helmet, CORS).

- Addresses real socio-economic challenges in Egypt by promoting financial literacy and helping users organize income, expenses, saving goals, and documents in one secure platform.

AI Insight Visualizer

React, TypeScript, Express, PostgreSQL

- Built a full-stack AI-powered analytics platform for dataset analysis and visualization.
- Designed a type-safe monorepo architecture using OpenAPI specifications and Zod validation.
- Implemented AI-generated insights, reporting, dataset management, and visualization recommendations.
- Integrated PostgreSQL, Redis, BullMQ, and OpenAI services.

Passport OCR & Document Verification System

FastAPI, EasyOCR, OpenCV

- Built an OCR platform for extracting and validating passport and ID document information.
- Implemented ICAO-compliant MRZ detection and parsing.
- Added document authenticity verification and facial quality analysis.
- Developed REST APIs and monitoring dashboards.

Hexaverse Authentication Backend

Node.js, Express.js, MongoDB

- Developed a secure authentication backend supporting Email/Password and Google OAuth.
- Implemented JWT authentication, password recovery, email verification, and rate limiting.
- Designed scalable APIs for mobile and web applications.

LoveDA Semi-Supervised Segmentation

PyTorch, U-Net

- Trained a semantic segmentation model using sparse point-level supervision.
- Developed a custom Partial Cross-Entropy loss function.
- Optimized model training for low-resource GPU environments.

Certifications

- AWS Certified AI Practitioner – Manara Learning Path
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI
- Understanding Data Science – DataCamp • Exploratory Data Analysis in Python – DataCamp
- CCNA: Introduction to Networks – Cisco Networking Academy
- Junior Cybersecurity Analyst Career Path – Cisco Networking Academy
- Cyber Threat Management – Cisco Networking Academy • Software Engineering Level 1 – Orange Digital Center
- CIB Summer Program – The Green Leap

Languages

Arabic (Native) | English (B2) | French (B2)